

1. AI-Powered Smart Study System

Problem Statement:

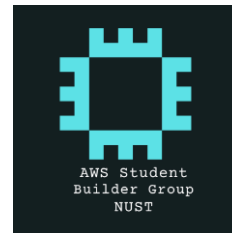
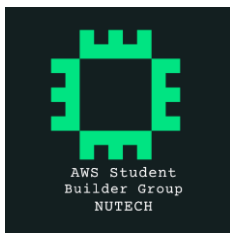
Students struggle with fragmented learning because educational content is scattered across books, notes, and online platforms. This results in poor concept clarity, inefficient revision, and lack of personalized practice.

System Requirements:

- The system must accept **user input in multiple forms**:
 - Text questions
 - Notes or paragraphs
 - Optional document content (PDF/text paste)
- The system must generate **multi-level explanations**:
 - Basic explanation (for beginners)
 - Intermediate conceptual breakdown
 - Advanced/technical explanation
- The system must generate **context-aware learning material**:
 - At least 5 quiz questions per topic
 - Mixed difficulty (easy, medium, hard)
- The system should maintain **session context**, meaning follow-up questions must relate to previous input.
- The system must ensure **accuracy-focused responses**, avoiding hallucinated or unrelated content.

Goal Behaviour:

A student should be able to input “Quantum Physics basics” and continuously interact with the system as if it is a structured tutor that adapts explanations and tests understanding.



2. AI Productivity & Time Optimization System

Problem Statement:

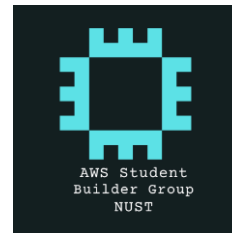
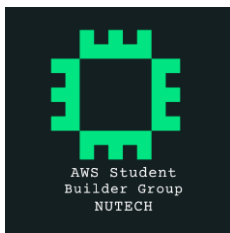
Individuals are unable to identify inefficiencies in their daily routines because raw activity logs do not translate into actionable insights.

System Requirements:

- The system must accept **structured or unstructured activity data** such as:
 - Text logs of daily tasks
 - Time-duration entries
- The system must automatically:
 - Categorize activities into meaningful groups (study, work, distraction, rest)
 - Detect **low-value time blocks**
- The system must generate:
 - A **daily efficiency report**
 - A list of **productivity leaks**
 - At least **3 actionable improvements**
- The system must generate a **suggested optimized schedule** based on past patterns.
- The system must support **trend detection over multiple entries** (even if simulated in hackathon).

Goal Behaviour:

If a user enters “2 hours YouTube, 1 hour study, 3 hours coding,” the system should clearly identify imbalance and suggest a better time distribution.



3. AI Landing Page Generator System

Problem Statement:

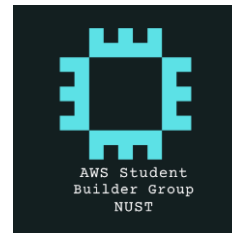
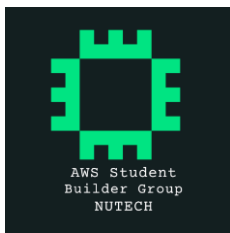
Users without technical or design knowledge cannot convert ideas into structured web presentations, delaying idea validation and product launches.

System Requirements:

- The system must accept a **natural language product description**:
 - Example: “AI tool for fitness tracking with diet suggestions”
- The system must generate a **structured landing page blueprint** containing:
 - Hero section (title + tagline)
 - Feature sections (at least 3–5 features)
 - Call-to-action logic
 - Optional FAQ section
- The system must ensure:
 - Logical structure consistency (sections must relate to input idea)
 - No missing required sections
- The system must output:
 - A **complete renderable page structure**
 - OR structured JSON that represents the page
- The system should allow **regeneration or refinement** of sections based on feedback.

Goal Behaviour:

A user should input a startup idea and receive a fully structured marketing page that feels like a real product landing page.



4. AI Customer Support System

Problem Statement:

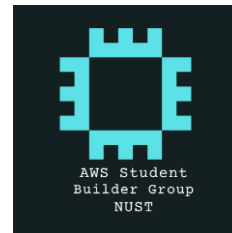
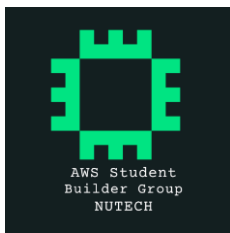
Small businesses cannot handle continuous customer queries, leading to delayed responses and loss of customers.

System Requirements:

- The system must allow ingestion of **business knowledge base**, such as:
 - FAQs
 - Product descriptions
 - Policies (refund, delivery, etc.)
- The system must answer user queries based **ONLY** on provided knowledge.
- If information is missing, the system must:
 - Clearly respond that it is unavailable
 - Suggest escalation or fallback response
- The system must support:
 - Multi-turn conversations
 - Context retention within a session
- The system must classify queries into:
 - Support, billing, product, or general info

Goal Behaviour:

If asked “How long does delivery take?”, it should only respond based on provided business data, not hallucinated information.



5. AI Resume & Job Matching System

Problem Statement:

Job seekers often apply to irrelevant jobs because they cannot accurately assess resume-job compatibility.

System Requirements:

- The system must accept:
 - Resume content (text or parsed format)
 - Job description text
- The system must compute:
 - **A match score (0–100%)**
 - Skill overlap analysis
- The system must identify:
 - Missing required skills
 - Weakly matched skills
- The system must generate:
 - Resume improvement suggestions
 - Optional rewritten bullet points for improvement
- The system must ensure:
 - Comparison is based on semantic understanding, not keyword matching only

Goal Behaviour:

A user should clearly understand why they are or are not a good fit for a job and what to improve.